

LLAMAINDEX

documentation helper llamaindex

Source: instructions.md

1. Create virtual environment

Create virtual environment in this directory using:

```
pipenv shell
```

2. Install dependencies

Install dependencies using:

```
pipenv install requests beautifulsoup4
```

3. Download documentation

Run the following command:

```
python download_docs.py
```

This downloads documentation from the provided URL.

This should download many `.html` files under:

```
llamaindex-docs/
```

(not just `sample.html`).

4. Install ingestion libraries

Run:

```
pipenv install python-dotenv llama-index
```

This installs required libraries for **local ingestion** (no paid APIs).

5. Run ingestion

Execute:

```
pipenv run python ingestion.py
```

This uses **local storage by default**:

```
INGEST_BACKEND=local
```

No **OpenAI** or **Pinecone keys** are required.

6. Optional Pinecone backend

If you explicitly want **Pinecone backend (without OpenAI)**, install extras:

```
pipenv install pinecone llama-index-vector-stores-pinecone
```

Then set in `.env`:

```
INGEST_BACKEND=pinecone
PINECONE_API_KEY
PINECONE_INDEX_NAME
```

Optional variables:

```
PINECONE_HOST
PINECONE_DIMENSION
```

Default dimension:

```
1536
```

7. Pinecone index configuration

If using Pinecone:

Create an index with metric:

```
cosine
```

The index dimension must match your **embedding dimension**:

```
PINECONE_DIMENSION
```

8. Run chat UI

Install Streamlit:

```
pipenv install streamlit
```

Run UI:

```
pipenv run streamlit run main.py
```

9. Pinecone retrieval backend

`main.py` defaults to **local retrieval backend** (no paid keys).

To explicitly use **Pinecone retrieval backend**, run:

```
QUERY_BACKEND=pinecone pipenv run streamlit run main.py
```

10. Text chunking parameters

Generally we use:

```
chunk size = 500  
chunk overlap = 50
```

for text splitting.

These values can be adjusted in:

```
ingestion.py
```

11. LlamaHub

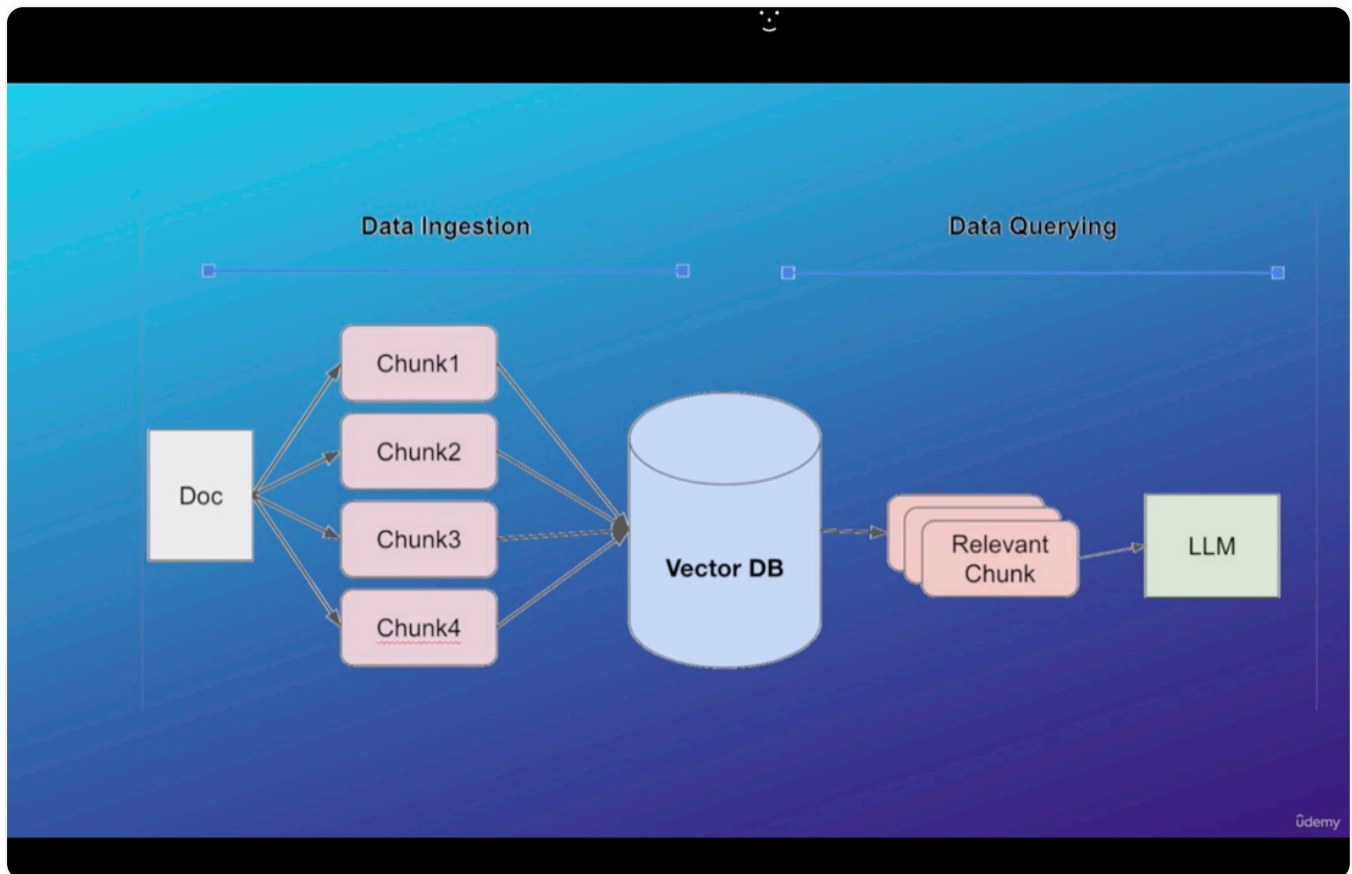
<https://llamahub.ai/>

12. Pinecone Vector Store

<https://app.pinecone.io/organizations/-OmuNosr5h78t5A7BUd8/projects/81d4ac51-5036-45e0-8ec3-97d8ba909ca3/indexes/llamaindex-documentation-helper/browser>

13. RAG pipeline

Find the below RAG pipeline which we are building here.



RAG Pipeline

14. Execution order

Run the following commands in sequence:

```
pipenv run python download_docs.py
pipenv run python ingestion.py
pipenv run streamlit run main.py
```

15. Retrieval Augmented Generation flow

LlamaIndex takes the **original prompt (query)** and feeds it to the **Pinecone vector store**.

Pinecone finds **k-nearest neighbor vectors** to the prompt.

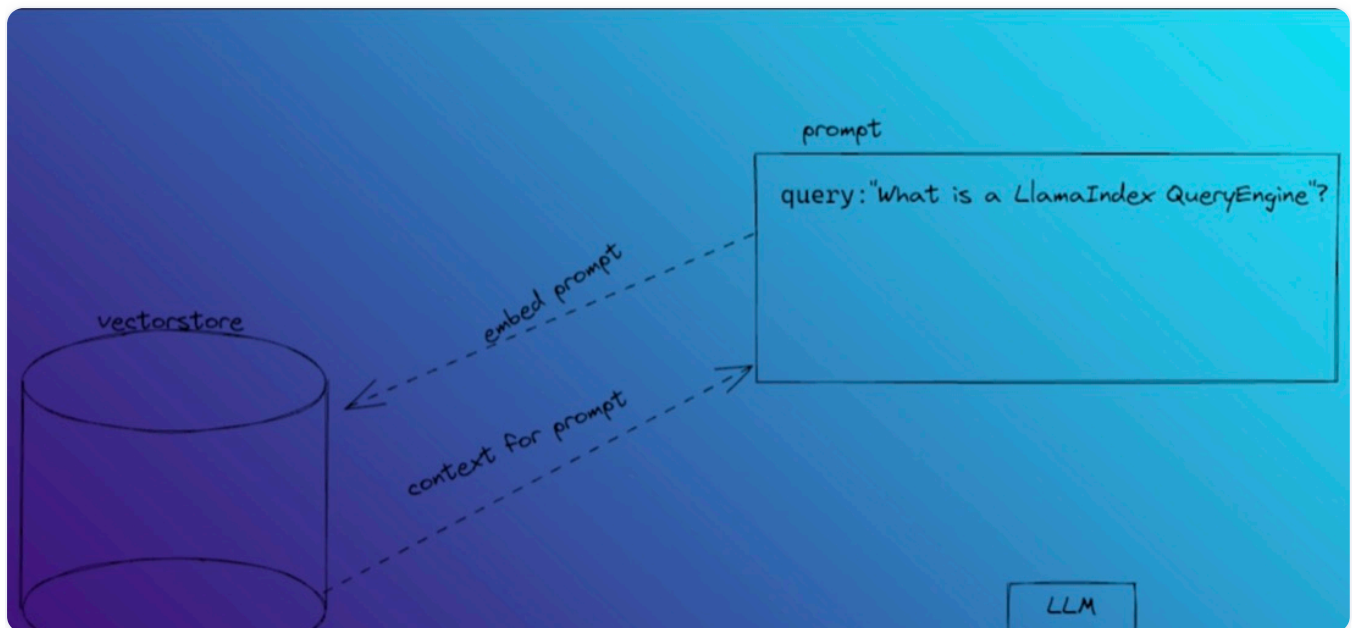
These neighbor vectors become the **context for the augmented prompt**.

This process is called:

- semantic search
- similarity search

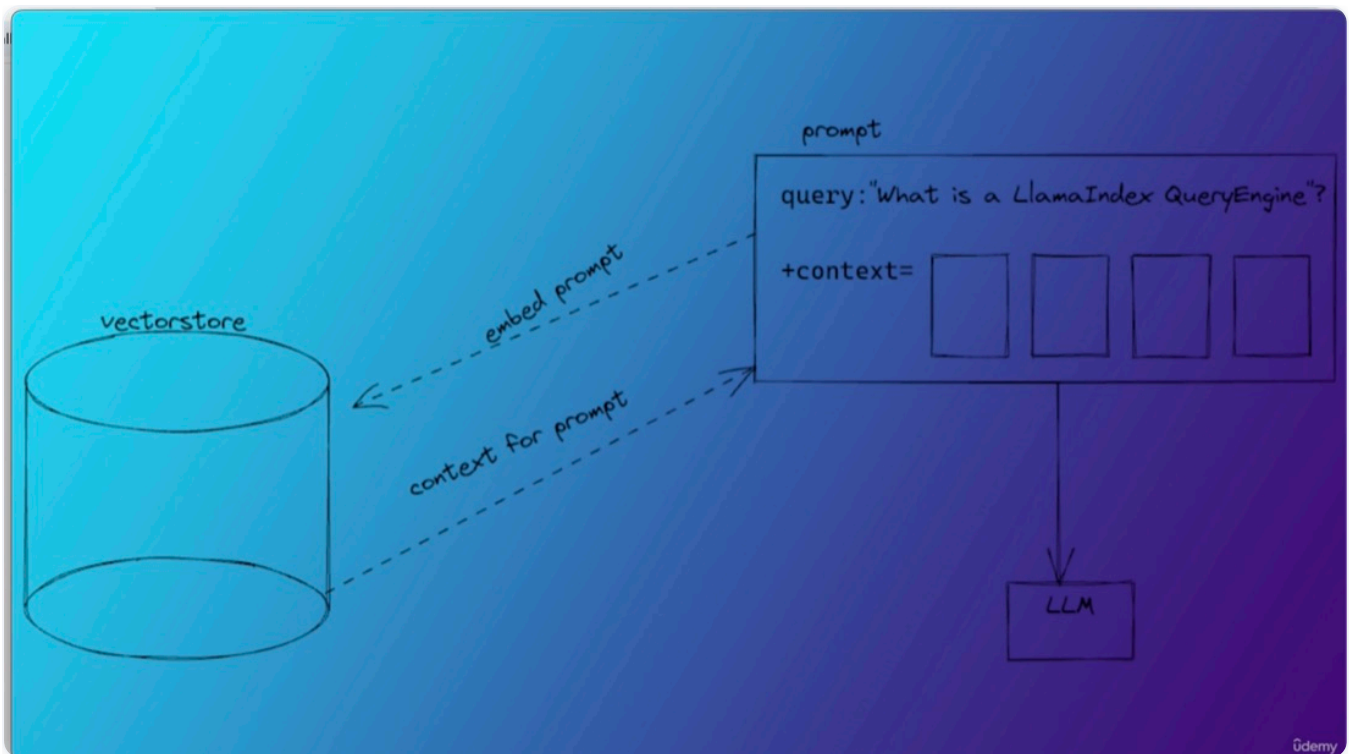
This is the **key principle of retrieval augmentation**.

Pinecone returns those vectors and these vectors become the **context vectors for the prompt**.



RAG Pipeline

After that, we take the **prompt and augment it with the context vectors**:



RAG Pipeline

16. Callbacks

Concept

LlamaIndex provides callbacks to help **debug, track, and trace** the inner workings of the library.

Using the **callback manager**, multiple callbacks can be added.

In addition to logging event data, you can also track:

- duration of events
- number of occurrences of events

A **trace map of events** is also recorded.

Callbacks can use this data however they want.

Example:

The **LlamaDebugHandler** prints the trace of events after most operations.

Callback Event Types

Available events that can be tracked:

- **CHUNKING** → Logs for before and after text splitting.
- **NODE_PARSING** → Logs for documents and nodes created from them.
- **EMBEDDING** → Logs for number of texts embedded.

- **LLM** → Logs for template and response of LLM calls.
 - **QUERY** → Tracks start and end of each query.
 - **RETRIEVE** → Logs for nodes retrieved for a query.
 - **SYNTHESIZE** → Logs results of synthesize calls.
 - **TREE** → Logs summaries and summary levels generated.
 - **SUB_QUESTION** → Logs generated sub-questions and answers.
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17. Streamlit

<https://docs.streamlit.io/get-started>

<https://github.com/streamlit/streamlit>

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